Blessed G/Michael Catholic Secondary School

Semester 1 Chemistry Mid1 Examination for Grade10

Academic Year: 2024 / 2018 E.C. Time Allowed: 1:30'

Name: _____ **Grade & Sec:** Grade10 _____ **R.No:** _____

I. Choose the best answer among the given alternatives.

____ **1.** What is the general molecular formula of saturated hydrocarbons (Alkanes)?
A. CnH2n **B.** CnH2n+2 **C.** CnH2n-2 **D.** CnH2n-6

____ **2.** What is the hybridization of carbon atoms in an alkane molecule like methane (CH4)?
A. sp **B.** sp2 **C.** sp3 **D.** dsp2

____ **3.** What is the IUPAC name for the hydrocarbon CH3-CH(CH3)-CH2-CH3?
A. 2-methylpropane **B.** 2-methylbutane
C. Isopentane **D.** Dimethylpropane

____ **4.** How many structural isomers does butane (C4H10) have?
A. 1 **B.** 2 **C.** 3 **D.** 4

____ **5.** Complete combustion of any hydrocarbon in excess oxygen produces:
A. CO and H2 **B.** CO2 and H2O **C.** C and H2O **D.** CO2 and H2

____ **6.** Which of the following hydrocarbons contains a triple bond between carbon atoms?
A. Ethene **B.** Ethane **C.** Ethyne **D.** Propene

____ **7.** The reaction between methane and chlorine in the presence of ultraviolet light is an example of:
A. Addition reaction **B.** Substitution reaction
C. Elimination reaction **D.** Polymerization

____ **8.** According to Markovnikov's rule, when HBr adds to propene (CH3-CH=CH2), the hydrogen atom attaches to:
A. The carbon with fewer hydrogen atoms **B.** The carbon with more hydrogen atoms
C. The central carbon exclusively **D.** Any carbon randomly

____ **9.** What type of isomerism is exhibited by but-2-ene in cis and trans arrangements?
A. Chain isomerism **B.** Positional isomerism
C. Geometric (stereoisomerism) **D.** Functional group isomerism

____ **10.** Which reagent is used as a test for unsaturation (decolorized by alkenes and alkynes)?
A. Bromine water (Br2 in CCl4) **B.** Dilute hydrochloric acid
C. Sodium hydroxide solution **D.** Litmus indicator

____ **11.** Which property of carbon allows it to form endless chains and rings of covalent bonds?
A. Catenation **B.** Allotropy
C. Electronegativity **D.** Radioactivity

____ **12.** The bond angles in ethene (C2H4) around each sp2-hybridized carbon atom are approximately:
A. 120 degrees **B.** 109.5 degrees **C.** 180 degrees **D.** 90 degrees

____ **13.** What is the principal component of natural gas (often exceeding 90% by volume)?
A. Methane (CH4) **B.** Propane (C3H8) **C.** Butane (C4H10) **D.** Ethyne (C2H2)

____ **14.** What is the IUPAC name of the compound CH3-C(CH3)2-CH3?
A. 2,2-dimethylpropane **B.** Neopentane
C. 2-methylbutane **D.** Tetramethylmethane

____ **15.** Fractional distillation of petroleum separates crude oil fractions based on differences in their:
A. Boiling points **B.** Densities
C. Solubility in water **D.** Chemical reactivity

____ **16.** Which of the following hydrocarbons burns with the sootiest, most luminous yellow flame due to high carbon content?

A. Ethyne (C2H2) **B.** Methane (CH4) **C.** Ethane (C2H6) **D.** Propane (C3H8)

____ **17.** Catalytic cracking in petroleum refining is primarily carried out to:

A. Convert long-chain heavy alkanes into shorter, high-demand gasoline fractions and alkenes
B. Extract sulfur impurities only
C. Synthesize coal
D. Cool down petroleum tanks

____ **18.** What is the functional group of alkenes?

A. Carbon-carbon double bond (C=C) **B.** Carbon-carbon triple bond

C. Hydroxyl group (-OH) **D.** Carbonyl group (>C=O)

____ **19.** Which gas is commonly used in commercial agriculture to artificially accelerate fruit ripening?

A. Ethene (ethylene, C2H4) **B.** Methane

C. Acetylene **D.** Carbon monoxide

____ **20.** What is the general molecular formula of alkynes?

A. CnH2n-2 **B.** CnH2n+2 **C.** CnH2n **D.** CnH2n-4

____ **21.** Reaction of calcium carbide (CaC2) with water produces which gas?

A. Ethyne (acetylene, C2H2) **B.** Methane

C. Ethene **D.** Hydrogen gas only

____ **22.** Hydrogenation of vegetable oils into solid margarine in the presence of nickel catalyst is an example of:

A. Catalytic addition reaction across C=C bonds
B. Substitution reaction
C. Polymerization
D. Oxidation

____ **23.** In fractional distillation of liquid air, which gas boils off first due to its lower boiling point (-196°C)?

A. Nitrogen (N2) **B.** Oxygen (O2, bp -183°C)

C. Argon (Ar, bp -186°C) **D.** Carbon dioxide

____ **24.** The major organic product formed when chlorine reacts with benzene in the presence of FeCl3 catalyst is:

A. Chlorobenzene **B.** Hexachlorocyclohexane

C. Benzyl chloride **D.** Phenol

____ **25.** Which test visually distinguishes terminal alkynes (like ethyne) from internal alkynes?

A. Formation of red precipitate with ammoniacal cuprous chloride Cu2Cl2
B. Bromine water decolorization
C. Litmus paper change
D. Combustion in air